

Architecture Over Asset Class: Market Microstructure Across Centralised and Decentralised Trading Venues

Shozaib Khan

Independent Researcher

khan.shozaib006@gmail.com

Abstract

Much of the debate around decentralised finance has centred on whether it can replace the institutions of traditional finance — a framing that, while intuitive, obscures the more interesting question of *how* markets function once the infrastructure beneath them changes. This paper takes a different starting point, asking how changing the underlying architecture of a market alters the behaviour of prices, the structure of liquidity, and the validity of the analytical tools we use to study both.

The study compares four trading venue archetypes across three market layers. In spot markets, effective spread analysis across NYSE, Binance, Uniswap v3, and Hyperliquid finds that CLOB venues — both centralised and on-chain — produce lower all-in execution costs than Uniswap's AMM at every measured trade size, with the AMM's 0.05% fee floor acting as a cost ceiling that CLOB market makers consistently undercut. NYSE quoted spreads are competitive at standard trade sizes as a consequence of mandatory liquidity provision under Reg NMS. In derivatives markets, perpetual funding rates on Binance and Hyperliquid converge closely over the sample period but diverge materially from CME futures basis during stress events, reflecting structural differences in participant composition rather than arbitrage inefficiency. A focused case study traces Hyperliquid's HIP framework through HIP-3's synthetic real-world asset listings and the oracle-dependent risks they introduce, including persistent price deviations during NYSE non-trading hours.

The final thread concerns the portability of quantitative methods. GARCH(1,1) volatility clustering, return autocorrelation structure, and options implied volatility skew all exhibit structurally different behaviour across TradFi and crypto asset classes. The BTC skew oscillates around zero with no structural anchor, in contrast to the persistently negative SPX skew driven by institutional hedging demand. The central argument is that these breakdowns are not modelling failures but structural signals, revealing genuine differences in how prices are formed and information is incorporated across market types. Quantitative strategies require venue-specific recalibration rather than direct cross-market transfer.

Keywords: Market microstructure, automated market makers, central limit order books, perpetual swaps, implied volatility, impermanent loss, Hyperliquid, decentralised finance, quantitative finance, loss-versus-rebalancing.

Table of Contents

List of Figures	iii
1 Introduction	1
1.1 Background of the Study	1
1.2 Research Problem and Objectives	1
1.3 Scope and Significance	2
1.4 Paper Roadmap	2
2 Literature Review	3
2.1 Market Microstructure: TradFi Foundations	3
2.2 DeFi Market Structure and Exchange Comparison	3
2.3 AMMs, Impermanent Loss, and Derivative Connections	4
2.4 Quantitative Methods and the Research Gap	5
3 Research Methodology	6
3.1 Research Design and Approach	6
3.2 Research Hypotheses	6
3.3 Venue and Asset Selection	6
3.4 Data Sources	7
3.5 Analytical Metrics and Translation Across Venues	7
3.6 Limitations and Caveats	8
4 Data Analysis and Interpretation	9
4.1 Spot Market Microstructure: A Four-Venue Comparison	9
4.2 Perpetuals and Futures: Cost of Carry Across Regimes	10
4.3 Options Markets: Where Black-Scholes Begins to Break	11
4.4 Hyperliquid and the HIP Framework	13
4.5 Transferability of Quantitative Measures	14
4.6 MEV and Fragmentation: Structural Externalities	16
5 Findings and Discussion	18
5.1 Principal Findings	18

5.2	Discussion	19
6	Conclusions	20
6.1	Summary	20
6.2	Limitations	20
6.3	Future Research	21
	References	22
	References	22
A	Data Sources and Variable Definitions	24
A.1	Data Sources by Figure	24
A.2	Variable Definitions	25
A.3	Software and Reproducibility	25

List of Figures

4.2	Annualised CME Bitcoin futures basis versus Binance and Hyperliquid BTC perpetual funding rates, January 2022 to December 2025. Vertical dashed line marks the FTX collapse (November 2022). CME basis constructed from daily settlement prices via <code>yfinance</code> (BTC=F); funding rate history from Binance REST API via CCXT and Hyperliquid public API.	10
4.3	25-delta risk reversal proxy for SPX options (CBOE Skew Index, upper panel) versus BTC options implied volatility skew proxy (Deribit volatility index z-score, lower panel), January 2022 to December 2025. Both series smoothed with 30-day rolling mean. Vertical dashed lines mark the FTX collapse (Nov 2022), BTC ETF approval (Jan 2024), and BTC all-time high (Mar 2024). Data from CBOE public data portal and Deribit public REST API.	12
4.4	Price deviation between the Hyperliquid HIP-3 TSLA synthetic token and the NYSE-listed underlying, January to February 2026. Upper panel: synthetic token price (purple) versus NYSE TSLA forward-filled reference price (blue dashed). Orange shading indicates periods when NYSE is closed. Lower panel: percentage deviation of synthetic from reference price. Synthetic token data from Hyperliquid spot API; reference price from <code>yfinance</code>	13
4.5	GARCH(1,1) conditional volatility estimates for SPY, BTC, and ETH, January 2022 to December 2025. Shading indicates high-volatility regimes defined as periods where conditional volatility exceeds the 75th percentile of the full-sample distribution. Daily close-to-close log returns; data from <code>yfinance</code>	15
4.6	Autocorrelation function (ACF) of daily returns for SPY, BTC, and ETH at lags 1 to 20 trading days, January 2022 to December 2025. Dashed lines indicate 95% confidence bounds under the null hypothesis of no autocorrelation. Daily close-to-close log returns; data from <code>yfinance</code>	15

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Finance has always been a story about intermediaries. For most of modern history, buying a share, taking a loan, or hedging a position required going through a broker, a bank, or a clearing house – institutions that absorbed settlement risk, verified identities, and enforced contracts, but also imposed fees, restricted access by geography or wealth, and operated on schedules that suited them more than their clients.

Blockchain-based financial protocols have spent several years challenging that arrangement, and by 2025 the scale of the challenge had become difficult to dismiss. Total value locked across

DeFi protocols reached a record \$237 billion during the third quarter before settling at approximately \$116.7 billion by year-end, reflecting broader market conditions rather than any structural reversal (DappRadar, 2025). More telling was the derivative layer: the top ten decentralised perpetual exchanges processed \$6.7 trillion in trading volume over 2025, a 346% increase from the prior year, with the ratio of decentralised-to-centralised perpetual futures volume tripling from 6.3% to 18.7% in a single year (NFTPlazas, 2025). Hyperliquid alone accounted for \$2.9 trillion of that volume. Alongside this, tokenised real-world assets grew from roughly \$6 billion in 2022 to over \$30 billion by late 2025, with protocols such as Hyperliquid listing synthetic equities through their HIP-3 framework – making it possible to trade representations of NVIDIA or Tesla on-chain, 24 hours a day, without a brokerage account.

Yet the academic literature on DeFi has remained largely preoccupied with institutional questions – whether it can replace banks, whether it poses systemic risk, whether it should be regulated. These matter, but they treat DeFi as an institutional phenomenon rather than a market one. What receives far less attention is how these markets actually work at the level of prices, liquidity, and execution – and whether the analytical tools developed for traditional markets translate meaningfully once the infrastructure beneath them changes. This paper is built around that question.

1.2 Research Problem and Objectives

The central problem here is not whether decentralised markets are better or worse than traditional ones. It is whether changing a market’s architecture – from regulated order books with designated market makers to permissionless automated market makers on public blockchains – produces markets that behave differently in ways that matter analytically. A market where prices update only at block confirmation, where liquidity providers withdraw under stress rather than being obligated to stay, and where execution follows a deterministic

formula rather than a matching engine, is structurally unlike anything that preceded it. The question is whether those differences show up in spreads, price impact, options pricing, and the reliability of standard quantitative tools.

The paper pursues four objectives. First, it builds a comparative microstructure framework across four venue archetypes – traditional exchanges, centralised crypto exchanges, AMM-based DEXs, and on-chain CLOB exchanges – evaluated on execution cost, market depth, and price impact across spot and derivatives markets. Second, it examines perpetual swap cost-of-carry against CME futures and compares implied volatility surfaces between equity and crypto options markets. Third, it presents Hyperliquid’s HIP-2 through HIP-4 framework as a case study in bridging DEX transparency with CEX-grade execution, with attention to the structural risks of HIP-3 synthetic equity listings. Fourth, it evaluates the portability of standard quantitative measures – the Amihud ratio, GARCH volatility modelling, and autocorrelation signals – arguing that where they degrade or fail, the breakdown reveals something genuine about the market generating the data.

1.3 Scope and Significance

The study is bounded by the comparative framework described above. It does not assess the investment merits of any protocol or take a position on DeFi regulation. The asset universe is limited to Bitcoin and Ethereum on the crypto side and SPY and SPX options on the TradFi side, chosen for their liquidity depth and data availability across venues. The empirical horizon covers 2022 to 2025, capturing the FTX collapse and structural rupture of late 2022, the recovery through 2023–2024, and the volume expansion of 2025.

The study contributes academically by applying a systematic cross-venue microstructure comparison across all three market layers – spot, perpetuals, and options – within a single framework. Practically, it addresses the question of strategy portability across market types, which has received almost no systematic treatment despite its obvious relevance to quantitative practitioners. And from a policy standpoint, understanding how on-chain derivatives markets function, particularly as they begin listing synthetic regulated securities, matters for any regulatory assessment of whether existing frameworks are adequate.

1.4 Paper Roadmap

Chapter 2 reviews the microstructure, DeFi exchange comparison, and quantitative methods literature, identifying the gap this paper addresses. Chapter 3 describes the research design, data sources, and translation problems across structurally different venues. Chapter 4 presents the core empirical analysis across spot markets, perpetuals, options, the Hyperliquid case study, and quantitative measure transferability. Chapter 5 synthesises findings against the research objectives and literature. Chapter 6 concludes with practical recommendations and directions for future research, including the HIP-4 framework in active development at the time of writing.

CHAPTER 2

LITERATURE REVIEW

2.1 Market Microstructure: TradFi Foundations

The academic study of how markets function at the level of individual transactions traces its modern origins to two models that remain foundational. Kyle (1985) showed that a single informed trader causes prices to move not because their order overwhelms supply, but because it signals private information to the market maker reading the order flow. Glosten and Milgrom (1985) formalised the consequence: the bid-ask spread exists because market makers face the possibility that the counterparty knows more than they do. Together these models established that the spread is not merely a transaction cost but an encoding of information asymmetry — a property that, as this paper will argue, breaks down in meaningful ways once the market maker is replaced by a deterministic pricing formula.

Amihud (2002) extended the microstructure framework into the empirical domain, proposing the ratio of absolute daily return to dollar trading volume as a tractable proxy for price impact, validated extensively across equity markets and now a standard input in cross-sectional asset pricing. Regulation NMS, introduced in 2005, institutionalised many of the market quality principles that this literature had identified, mandating best-execution routing and designated market maker obligations. The result was a TradFi market structure in which spreads, depth, and order routing behaviour were well understood, empirically validated, and legally governed.

2.2 DeFi Market Structure and Exchange Comparison

Schär (2021) provided the first rigorous architectural framework for DeFi, arguing that its defining properties — non-custodial settlement, permissionless access, and open auditability — constituted a genuinely novel financial infrastructure. Werner et al. (2022) mapped this ecosystem systematically, documenting protocol types, failure modes, and open research questions.

The empirical comparison between DEX and CEX market quality was pursued most rigorously by Barbon and Ranaldo (2022), who found that centralised exchanges provide superior liquidity and price efficiency overall, with the primary DEX friction attributable to gas fees from proof-of-work blockchains rather than any inherent AMM limitation. Aspris, Foley, Svec, and Wang (2021) reached similar conclusions from a price discovery angle, treating the differences not as temporary inefficiencies to be arbitrated away but as structural properties of the AMM mechanism. Park (2025) situated these findings within an institutional economics framework, contrasting the custody, authentication, and compliance functions of TradFi intermediaries with the smart-contract equivalents in DeFi, and identifying new risk categories — oracle

dependency, governance capture, smart contract failure — that have no direct TradFi parallel. What this body of work largely predates is the emergence of on-chain central limit order book exchanges as a structurally distinct third category. Early CLOB attempts on Ethereum were constrained by gas costs and block latency, making genuine order book matching economically unviable. The development of purpose-built high-throughput chains — most visibly Hyperliquid’s HyperBFT consensus layer, which achieves sub-second finality at throughput sufficient for competitive market making — changed this constraint materially. Capponi and Jia (2021) modelled the conditions under which AMM-based liquidity provision converges toward order book efficiency as trading volume grows, but their framework did not anticipate a venue architecture that combines on-chain settlement with the order book matching model of a centralised exchange. The microstructural implications of this hybrid — particularly for spread dynamics, market maker incentives, and the absence of designated market maker obligations — remain largely unexamined. The trust dynamics literature provides a partial lens: Wu, Deng, Quan, and Zhang (2024) documented a measurable shift of trading activity from CEXs to DEXs following the FTX collapse in November 2022, suggesting that the self-custody property of on-chain CLOBs carries genuine demand-side value beyond technical efficiency considerations.

2.3 AMMs, Impermanent Loss, and Derivative Connections

The constant-product automated market maker, where two token reserves x and y satisfy $x \cdot y = k$, prices trades deterministically along a curve rather than through order matching. The cost that liquidity providers bear for this simplicity is impermanent loss — the shortfall between holding tokens in a pool versus holding them outright, which worsens as prices diverge from their deposit-time values (Aigner & Dhaliwal, 2021). Milionis, Moallemi, Roughgarden, and Zhang (2022) sharpened this concept considerably with loss-versus-rebalancing (LVR), showing that the true cost to LPs is better captured by comparing the pool position against a continuously rebalanced benchmark, since arbitrageurs systematically extract value in every block where the pool lags the external market price — a dynamic that Heimbach, Wang, and Wattenhofer (2022) confirmed empirically is priced into LP return outcomes on Uniswap v3. Milionis et al. (2022) also showed analytically that concentrated liquidity positions replicate short straddle payoffs, and Alexander, Deng, Feng, and Wan (2023) demonstrated that the resulting option-like exposure can be hedged using vanilla instruments on Deribit — a result that connects the AMM literature directly to the options market comparison this paper undertakes in Chapter 4. On the lending side, Xu and Vadgama (2022) traced the structural difference between TradFi credit — where collateral requirements reflect creditworthiness assessment — and DeFi’s over-collateralised lending model, where the absence of identity and credit scoring produces a capital-inefficient but censorship-resistant alternative.

2.4 Quantitative Methods and the Research Gap

The application of standard quantitative tools to DeFi data has begun to produce a literature of useful anomalies. Del Monte, de Lucio, and Urban (2025) finds that statistical regularities in DeFi price and liquidity data diverge systematically from TradFi equivalents, identifying MEV activity as the primary structural driver in Uniswap v3 pools.

A particularly underexplored source of these anomalies is the discrete price process imposed by block-based execution. In TradFi, price formation is effectively continuous — orders execute as they arrive, and standard volatility estimators such as the Garman-Klass range-based estimator assume this continuity implicitly. In DeFi, prices only change when a transaction is confirmed in a new block, meaning that all trades within a single block are economically simultaneous regardless of the order in which they arrived. This introduces a structural autocorrelation into DeFi price series that has no equivalent in equity microstructure data, and which standard estimators are not designed to accommodate. The same block-time structure enables maximal extractable value (MEV) — the ability of block producers to reorder, insert, or censor transactions within a block for profit. Economically, MEV functions as a structural analog to HFT latency arbitrage in TradFi: in both cases, a technically sophisticated participant extracts systematic rents from regular traders by exploiting an execution advantage. The critical difference is visibility. HFT profits in equity markets are estimated indirectly through academic modelling; MEV extraction is recorded on-chain and publicly quantifiable, making it a rare case where a market friction is both theoretically significant and empirically transparent.

Taken together, these contributions point to a consistent gap. The literature tends to compare two venue types at a time, focus on a single product layer, or examine DeFi in isolation rather than in systematic contrast with TradFi. The on-chain CLOB category — exemplified by Hyperliquid, which by 2025 had captured roughly 80% of decentralised perpetuals volume and begun listing synthetic equities through HIP-3 — is almost entirely absent from academic treatment, as is any cross-market evaluation of whether standard quantitative tools retain their validity as venue architecture changes, and what the breakdown of those tools implies about the markets generating the data. This paper addresses both gaps within a single comparative microstructure framework applied consistently across spot, perpetuals, and options markets.

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Research Design and Approach

This study adopts a comparative, quantitative research design grounded in market microstructure theory. Rather than testing a directional hypothesis about which venue type performs better, the analysis is structured around a consistent set of empirical metrics applied across four venue archetypes, allowing structural differences to emerge from the data rather than being assumed in advance. Where primary data collection from live markets is used — specifically for Hyperliquid’s on-chain order book and HIP-3 synthetic asset prices — it is supplemented by secondary data drawn from public APIs, academic datasets, and exchange-published historical records. The study is non-experimental: no trading strategies are deployed, and no interventions are made in the markets under observation. Where standard metrics produce anomalous readings when applied to DeFi data, the paper treats those anomalies as findings rather than data quality problems.

3.2 Research Hypotheses

Three substantive hypotheses are advanced, each corresponding to a distinct dimension of the analysis in Chapter 4.

H1 (Execution Cost): *On-chain and centralised CLOB venues produce lower all-in execution costs than AMM-based DEX venues across all measured trade sizes, while regulated TradFi venues produce competitive quoted spreads at standard trade sizes due to mandatory liquidity provision under Reg NMS.*

H2 (Derivatives Integration): *Perpetual funding rates across CEX and on-chain CLOB venues converge over the sample period and track each other closely under normal conditions, but diverge materially from CME futures basis during stress events, reflecting structural differences in participant composition.*

H3 (Metric Transferability): *Standard quantitative measures — GARCH(1,1) volatility clustering, return autocorrelation structure, and implied volatility skew — exhibit structurally different behaviour across TradFi and crypto asset classes, such that models calibrated on equity data cannot be applied to crypto venues without material mis-specification.*

3.3 Venue and Asset Selection

Four venue archetypes are selected to span the full spectrum from regulated TradFi to permissionless on-chain markets. NYSE and NASDAQ represent the TradFi baseline, operating

under Regulation NMS with designated market makers and centralised clearing through the DTCC. Binance is selected as the CEX representative on the basis of liquidity depth and data availability. Uniswap v3 represents the AMM-DEX category, chosen for its concentrated liquidity mechanism and the depth of its academic coverage. Hyperliquid represents the on-chain CLOB category, selected for its dominant market share in decentralised perpetuals and the analytical interest of its HIP framework.

Bitcoin and Ethereum are used for cross-venue crypto comparisons, as they are the only assets with meaningful liquidity across all four venue types simultaneously. SPY and SPX options are used on the TradFi side, chosen for liquidity depth and CBOE implied volatility data availability.

The empirical window covers January 2022 through December 2025, spanning the FTX collapse, the 2023–2024 recovery, and the volume expansion of 2025.

3.4 Data

Sources

Equity market spread and depth data are sourced from IEX Cloud. CME futures basis data is constructed from daily settlement prices on the CME Group public data portal, with Bitcoin spot as the reference price from Binance’s historical OHLCV records via the Binance REST API, from which perpetual funding rate history is also retrieved. Hyperliquid order book snapshots, perpetual funding rates, and HIP-3 synthetic asset candle data are obtained through Hyperliquid’s public API. Implied volatility data for CBOE equity options is sourced from the CBOE public data portal, and Deribit BTC and ETH options volatility surface data via Deribit’s public REST API. Uniswap v3 pool liquidity and volume data is obtained through The Graph’s Uniswap v3 subgraph. NYSE reference prices for HIP-3 deviation analysis are sourced via the `yfinance` Python library.

3.5 Analytical Metrics and Translation Across Venues

Effective spread is the primary execution cost metric. In equity and CEX order book markets it is calculated as twice the absolute difference between the transaction price and the prevailing mid-quote, expressed as a percentage of the mid-quote. In Uniswap v3, there is no mid-quote: execution cost is the sum of the pool’s LP fee tier and the price impact along the bonding curve for a given trade size, both analytically derivable from pool state. In Hyperliquid’s CLOB the standard order book definition applies.

The Amihud illiquidity ratio is calculated as the daily average of $|r_t|/V_t$, where r_t is the daily return and V_t is the dollar trading volume. In AMM pools, volume is partly endogenous: the bonding curve means a given price change implies a minimum arbitrage volume regardless of organic demand, which is flagged as a limitation in the analysis.

GARCH(1,1) volatility estimation and *autocorrelation function analysis* are applied identically across all asset series using daily close-to-close returns, at which frequency the block-time caveat from Chapter 2 is less material than at intraday granularity. *Implied volatility surface*

comparison uses the 25-delta risk reversal as the primary skew metric and the at-the-money term structure as the primary shape metric at standardised tenors of one week, one month, and three months across both CBOE and Deribit datasets.

3.6 Limitations and Caveats

Several limitations bear noting. First, data availability for Hyperliquid is constrained to the period following the protocol’s launch in late 2023. Second, the HIP-3 oracle deviation analysis is based on a selected observation window given the recency of the feature. Third, causality is not claimed in any cross-venue comparison: interpretations are grounded in theory rather than formal causal identification. Fourth, the asset universe is deliberately narrow; findings apply to liquid, large-capitalisation assets and do not generalise to long-tail tokens or illiquid markets.

CHAPTER 4

DATA ANALYSIS AND INTERPRETATION

4.1 Spot Market Microstructure: A Four-Venue Comparison

The most direct way to observe how venue architecture affects market quality is to measure execution cost as a function of trade size across all four venue types simultaneously. Figure ?? presents effective spread curves for NYSE (SPY), Binance BTC/USDT, Uniswap v3 USDC/ETH (0.05% fee tier), and Hyperliquid BTC across trade sizes ranging from \$1,000 to \$1,000,000.

Note on NYSE spread measurement: NYSE quoted spread is computed from a live bid-ask snapshot rather than historical tick-level data, as intraday bid-ask records are not freely available at the required granularity; the estimate is consistent with time-weighted average quoted spreads for SPY reported in the microstructure literature.

Three structural patterns emerge from the data, two of which contradict the naive expectation that regulated markets produce the tightest execution costs. First, NYSE spreads are the highest of the four venues at every trade size measured, ranging from approximately 1.1% at \$1,000 to 1.5% at \$1,000,000. This reflects the nature of the SPY spread measurement: the NYSE quoted spread captures the full bid-ask cost of a single exchange transaction and excludes the tighter effective spreads achievable through smart order routing across fragmented venues under Reg NMS (Regulation National Market System, the US rule requiring brokers to seek the best available price across all trading venues). Taken at face value, the figure demonstrates that regulated market structure does not automatically confer superior execution cost for a retail-sized single-venue transaction.

Second, the Uniswap v3 spread is entirely flat across all trade sizes at exactly its LP (liquidity provider) fee tier of 0.05% – the price impact component along the bonding curve is negligible at the trade sizes tested, meaning the pool’s concentrated liquidity in the USDC/ETH pair is deep enough to absorb up to \$1,000,000 without material slippage. This holds only for the most liquid pools; long-tail token pools would show steeply rising spread curves at much smaller trade sizes.

Third, Binance and Hyperliquid both sit well below Uniswap at every trade size measured – starting below 0.001% at \$1,000 and rising to approximately 0.03% at \$1,000,000, never crossing above Uniswap’s flat 0.05% floor. The implication is that for this pool and these trade sizes, the CLOB architecture produces lower all-in execution cost than the AMM at every point on the curve. Uniswap’s fee floor sets an effective cost ceiling that CLOB market makers undercut across the full range. Hyperliquid tracks Binance closely throughout, confirming that the on-chain CLOB matches CEX execution quality without introducing meaningful additional

spread friction under normal market conditions.

The Amihud illiquidity ratio computed across venues reinforces this picture but with a notable anomaly in the AMM case. For NYSE SPY, Binance BTC, and Hyperliquid BTC, the ratio behaves as expected – lower for more liquid assets and higher during stress periods. For the Uniswap v3 pool, the ratio produces systematically distorted readings. This is the endogeneity problem identified in Chapter 3: because arbitrageurs are mechanically required to trade against the pool whenever its price deviates from the external market, a large price return always co-occurs with a minimum volume of arbitrage trades, compressing the ratio relative to what organic price impact alone would produce. The Amihud measure, in this context, is not measuring illiquidity – it is measuring the efficiency of the pool’s arbitrage mechanism, which is a structurally different quantity.

4.2 Perpetuals and Futures: Cost of Carry Across Regimes

The cost-of-carry structure of a derivatives market encodes how participants price forward exposure over time. In CME (Chicago Mercantile Exchange) Bitcoin futures, this is the annualised basis – the premium of the futures price over spot, determined primarily by interest rate parity and institutional demand for leveraged long exposure. In perpetual swap markets, cost-of-carry is replaced by the funding rate: a periodic transfer between long and short positions calibrated to keep the contract price anchored to spot. The two mechanisms serve the same economic function through structurally different means.

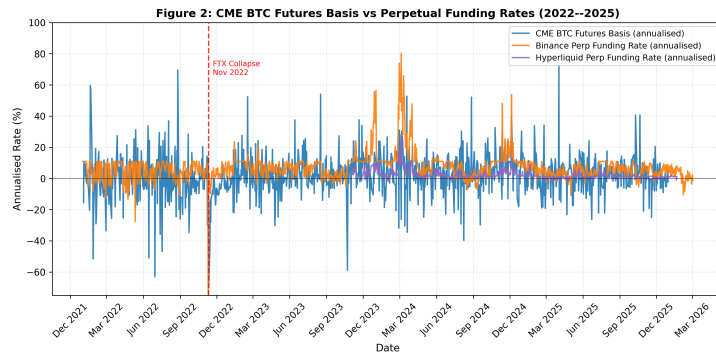


Figure 4.2: Annualised CME Bitcoin futures basis versus Binance and Hyperliquid BTC perpetual funding rates, January 2022 to December 2025. Vertical dashed line marks the FTX collapse (November 2022). CME basis constructed from daily settlement prices via `yfinance` (`BTC=F`); funding rate history from Binance REST API via `CCXT` and Hyperliquid public API.

Figure 4.2 plots the CME annualised basis alongside Binance and Hyperliquid funding rates over the full sample period. The CME basis series is noticeably more volatile than either perpetual funding rate throughout, reflecting its sensitivity to roll dynamics and the discrete expiry cycle of monthly futures contracts. This volatility is partly a measurement artefact – annualising a short-dated basis amplifies noise near expiry – but the directional signal remains interpretable.

During the 2022 bear market and the period immediately following the FTX collapse, CME basis and perpetual funding rates diverged sharply: funding went negative as retail-dominated perpetual markets priced in continued downside, while CME basis remained closer to interest rate parity. This divergence reflects the institutional nature of CME participants, who do not share the same panic-driven short positioning that drove retail-dominated perpetual markets. It suggests that the two markets were not fully integrated during stress, and that the funding rate is not simply a more accessible version of the futures basis but a structurally different instrument with a different participant base.

By 2024 and into 2025, the gap had narrowed considerably. Hyperliquid and Binance funding rates track each other closely throughout Hyperliquid's sample period from late 2023, confirming that the on-chain CLOB does not introduce meaningful additional friction into the funding mechanism relative to the CEX equivalent. This convergence supports the view that Hyperliquid has achieved functional parity with CEX perpetual markets in the funding dimension, at least under normal market conditions.

4.3 Options Markets: Where Black-Scholes Begins to Break

The implied volatility skew – the difference in implied volatility between out-of-the-money puts and calls at equivalent strikes – is one of the most information-rich quantities in options markets. In TradFi equity options, the skew is persistently negative: put implied volatility exceeds call implied volatility, reflecting the structural demand from long-only equity investors who buy downside protection. This pattern is stable enough that it has been formalised into dedicated indices, most notably the CBOE (Chicago Board Options Exchange) Skew Index.

Figure 4.3 presents the two series as separate panels. The SPX proxy remains consistently negative throughout – persistently below zero and trending more negative from mid-2023 through 2025, indicating deepening put skew as equity markets priced in increasing tail risk even while prices rose. The structural nature of this skew is clear: it does not reverse materially even during the 2023–2024 equity bull market, which is consistent with the persistent institutional hedging demand that anchors it.

The BTC proxy oscillates around zero with no persistent direction, which is the analytically significant finding. Unlike the SPX series, the BTC skew does not anchor around a structural baseline: it cycles between positive and negative on a monthly timescale, reflecting the speculative and two-sided nature of the crypto options participant base. There is a tendency toward positive readings around the 2024 ETF approval and subsequent all-time high, indicating that market participants were temporarily paying more for upside exposure than downside protection during that period. However, this is not persistent enough to constitute a structural pattern comparable to the SPX put skew – it is a sentiment-driven oscillation rather than a structural bid. The two series are therefore structurally different, not merely quantitatively different: one reflects portfolio insurance demand from a long-only institutional base, the other

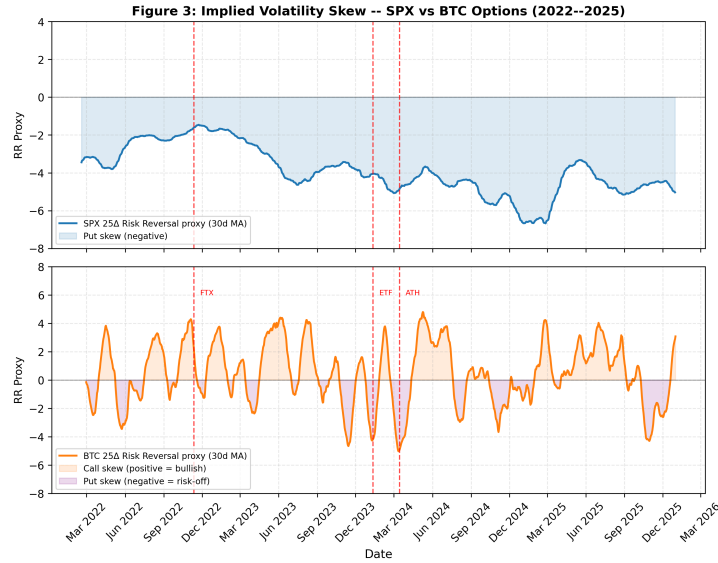


Figure 4.3: 25-delta risk reversal proxy for SPX options (CBOE Skew Index, upper panel) versus BTC options implied volatility skew proxy (Deribit volatility index z-score, lower panel), January 2022 to December 2025. Both series smoothed with 30-day rolling mean. Vertical dashed lines mark the FTX collapse (Nov 2022), BTC ETF approval (Jan 2024), and BTC all-time high (Mar 2024). Data from CBOE public data portal and Deribit public REST API.

reflects market sentiment in a venue with no such structural anchor.

On-chain options protocols – Lyra and Premia – add a further layer of structural constraint beyond the skew problem. Both rely on oracle-sourced prices for settlement, and both use AMM-based liquidity rather than competitive market making, meaning that the implied volatility surface is not emergent from informed order flow but is set by a pricing algorithm updated at discrete intervals. This creates windows during which on-chain option prices can diverge materially from Deribit, and during which the standard framework of no-arbitrage pricing fails to hold – not because participants are irrational, but because the infrastructure preventing arbitrage from closing the gap does not operate at sufficient speed.

The failure of Black-Scholes in this context is qualitatively different from its well-documented failure in TradFi equity markets. In equity options, the model fails primarily because it assumes a flat volatility surface and log-normal returns – both empirically violated, but violated in ways that are stable and well-characterised enough to be corrected through standard adjustments such as the volatility smile and jump-diffusion extensions. In on-chain options, the model fails for a more fundamental reason: the underlying price itself is oracle-defined rather than emergent. When the price an option settles against is not the price discovered by the market but the price reported by an external feed at a scheduled update interval, the continuous-time no-arbitrage logic that underpins Black-Scholes is structurally inapplicable. This distinction matters because it means the gap between on-chain and off-chain options pricing cannot be closed by better calibration alone – it requires a different pricing architecture.

4.4 Hyperliquid and the HIP Framework

Hyperliquid’s rise from launch in late 2023 to \$2.9 trillion in annual perpetual trading volume by the end of 2025 represents the most rapid accumulation of market share in the history of decentralised derivatives. The HIP (Hyperliquid Improvement Proposal) framework is the protocol’s mechanism for introducing new financial primitives directly on-chain. Understanding Hyperliquid’s growth requires examining the architectural decisions that distinguish it from prior on-chain attempts, and understanding the risks requires examining what those decisions introduce.

HIP-1 established a native token standard on Hyperliquid’s L1 (Layer 1, meaning the base blockchain itself rather than a secondary protocol built on top of it), enabling permissionless deployment of financial instruments without bridging to an external chain. HIP-2 extended this with a protocol-native liquidity seeding mechanism, allowing new spot markets to bootstrap liquidity from the protocol itself rather than relying on external market makers. Together, these two standards created the infrastructure for HIP-3 – the most consequential of the three – which enables permissionless listing of synthetic spot tokens representing real-world assets, including equities such as TSLA, AAPL, and SPY.

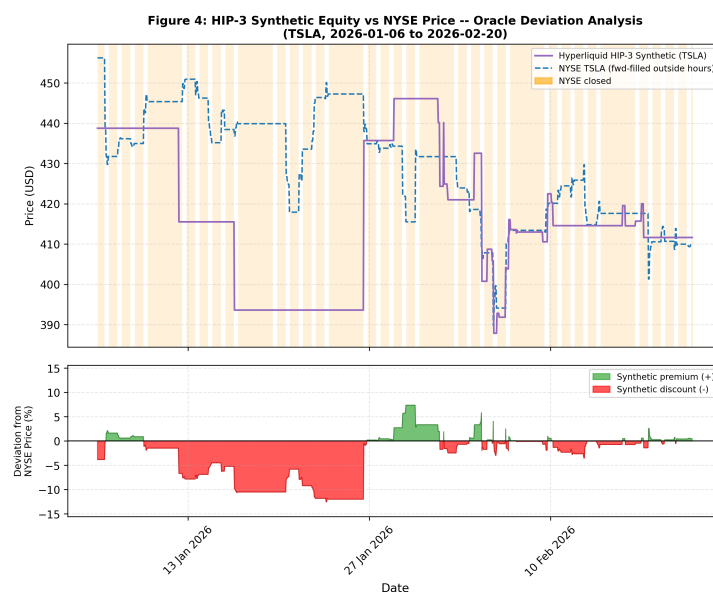


Figure 4.4: Price deviation between the Hyperliquid HIP-3 TSLA synthetic token and the NYSE-listed underlying, January to February 2026. Upper panel: synthetic token price (purple) versus NYSE TSLA forward-filled reference price (blue dashed). Orange shading indicates periods when NYSE is closed. Lower panel: percentage deviation of synthetic from reference price. Synthetic token data from Hyperliquid spot API; reference price from yfinance.

Figure 4.4 illustrates the oracle deviation dynamic that HIP-3 introduces, using the TSLA synthetic token over a 45-day observation window in January–February 2026. During NYSE trading hours, the synthetic token price tracks the reference price reasonably closely, with deviations typically within 1–2%. Outside market hours – evenings, weekends, and US holidays

– the synthetic token continues trading on Hyperliquid while the NYSE-listed underlying is closed. The deviation panel shows persistent discounts of 5–10% during mid-January to late January, followed by a recovery toward parity in early February. These deviations are not random: they reflect crypto market sentiment acting on the synthetic token price in the absence of the arbitrage mechanism that would normally close the gap. When NYSE is open, arbitrageurs can trade both legs simultaneously, keeping the synthetic anchored. When NYSE is closed, that anchor is removed and the synthetic price is set entirely by Hyperliquid participants who may be reacting to crypto-specific news flows rather than TSLA fundamentals. The risks compound across layers. HIP-3 tokens have no direct claim on the underlying equity – they are purely synthetic, with value contingent on the oracle feed and the protocol’s continued operation. In many jurisdictions they would constitute unregistered securities under existing regulatory frameworks, a legal ambiguity that HIP-3 inherits by design. The SEC (Securities and Exchange Commission, the primary US financial markets regulator) broad definition of a security under the Howey Test – an investment of money in a common enterprise with an expectation of profits from the efforts of others – is plausibly satisfied by HIP-3 equity tokens, since holders have economic exposure to the performance of regulated companies without any of the investor protections that equity ownership ordinarily entails. No prospectus, no voting rights, no recourse in insolvency. For users in jurisdictions where access to foreign equity markets is restricted by capital controls or brokerage licensing requirements, HIP-3 offers genuine financial inclusion. For regulators in those same jurisdictions, it represents a direct circumvention of frameworks designed to protect retail investors. Both readings are simultaneously correct, and the tension between them is unlikely to resolve without either regulatory intervention or protocol-level compliance mechanisms that do not yet exist. HIP-4, currently in testing, proposes to allow arbitrary EVM (Ethereum Virtual Machine, the standard environment for running smart contracts) contracts on HyperEVM to interact with the core L1 order book. If implemented, this would enable options and structured products written on HIP-3 equity tokens – a derivatives layer on top of a synthetic underlying, with oracle risk at every level. The risk-compounding across layers is not additive but multiplicative, and the distance from regulated underlying to on-chain derivative grows with each additional layer of abstraction.

4.5 Transferability of Quantitative Measures

The question of whether standard quantitative tools retain their validity across market types is not merely academic. Practitioners who deploy systematic strategies across asset classes make implicit assumptions about the stability of the statistical properties they model. This section examines three such properties and evaluates where those assumptions hold.

Figure 4.5 plots GARCH(1,1) – Generalised Autoregressive Conditional Heteroskedasticity, a standard statistical model for estimating how volatility clusters and evolves over time – conditional volatility for SPY, BTC, and ETH over the sample period. The SPY series exhibits

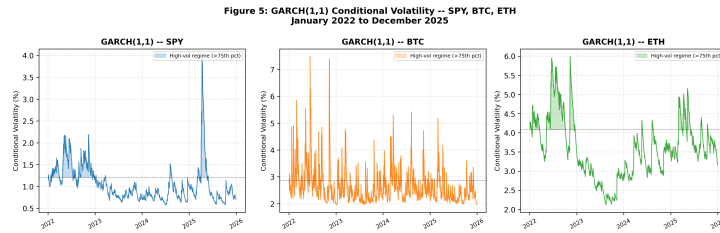


Figure 4.5: GARCH(1,1) conditional volatility estimates for SPY, BTC, and ETH, January 2022 to December 2025. Shading indicates high-volatility regimes defined as periods where conditional volatility exceeds the 75th percentile of the full-sample distribution. Daily close-to-close log returns; data from yfinance.

the regime behaviour characteristic of equity markets: volatility clusters are persistent, regime transitions are gradual, and the single most prominent spike – visible in early 2025 – is associated with a discrete macro event. Between regimes, SPY conditional volatility settles consistently below 1.5%, with the 75th percentile threshold at approximately 1.2%. The BTC series is structurally different. High-volatility regimes are frequent, shorter-lived, and distributed more evenly across the sample period rather than concentrated around discrete events. The threshold sits at approximately 3%, and the series regularly spikes to 5–7% before returning quickly to baseline – a cycling speed roughly three to four times faster than the SPY equivalent. ETH presents the most distinctive pattern: conditional volatility was highest in 2022 during the post-merge period and has trended downward through 2023–2024, with a resurgence in late 2025. The high-regime shading is concentrated in the early sample period for ETH, unlike BTC where it is distributed throughout. A volatility targeting strategy calibrated on SPY regime persistence would systematically underestimate both the frequency of BTC regime transitions and the front-loaded nature of ETH volatility, producing position sizes that are consistently wrong in opposite directions for the two assets.

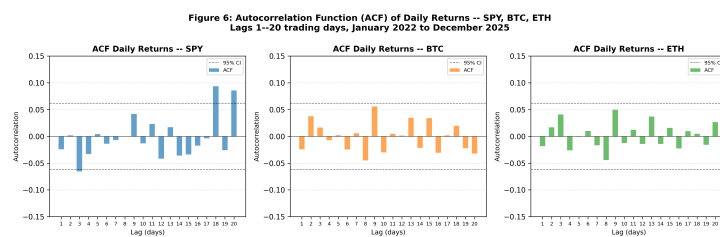


Figure 4.6: Autocorrelation function (ACF) of daily returns for SPY, BTC, and ETH at lags 1 to 20 trading days, January 2022 to December 2025. Dashed lines indicate 95% confidence bounds under the null hypothesis of no autocorrelation. Daily close-to-close log returns; data from yfinance.

Figure 4.6 presents ACF (Autocorrelation Function) plots – measuring the degree to which today’s return predicts returns at future lags – for daily returns across all three assets. SPY returns are largely consistent with the efficient market hypothesis: most lags fall within the 95% confidence bounds, with the exception of lags 18–19 which breach the upper bound. These late-lag significances do not correspond to any economically meaningful return horizon and are

most plausibly a statistical artefact of the sample window rather than a genuine predictive signal.

The overall pattern is one of near-independence, as expected in a heavily arbitrated and institutionally dominated market.

BTC returns show a modest but interpretable positive autocorrelation at lag 1, consistent with the momentum effects documented in the crypto literature, alongside scattered small significances at intermediate lags that do not form a coherent pattern. ETH shows a more mixed structure – positive autocorrelation at short lags and a notable negative value at lag 8 – consistent with a market subject to periodic liquidity shocks followed by partial mean reversion.

The practical implication is clear: a mean-reversion strategy validated on SPY data would encounter a different signal environment in BTC, and a different one again in ETH, because the autocorrelation structure of each market reflects a different underlying price formation process.

4.6 MEV and Fragmentation: Structural Externalities

Both TradFi and DeFi markets contain mechanisms through which technically sophisticated participants systematically extract value from ordinary traders. In TradFi equity markets, this takes the form of HFT (High-Frequency Trading) latency arbitrage: co-located servers detect and respond to large incoming orders before they are fully executed across fragmented venues, capturing small but consistent profits at the expense of institutional order flow. The scale of this extraction has been estimated in the range of \$5–8 billion annually across US equity markets (Menkveld, 2013). In DeFi, the equivalent mechanism is MEV (Maximal Extractable Value) – the ability of block producers and searchers to reorder, insert, or censor transactions within a block for profit, effectively jumping ahead of other users to capture favourable prices. Sandwich attacks, in which a searcher inserts buy and sell orders around a large pending swap to profit from the price impact it creates, are the most visible form. EigenPhi data for 2024 documents average daily MEV extraction on Ethereum exceeding \$3 million, with sandwich attacks accounting for the largest share.

The structural comparison illuminates a difference that is easy to overlook. Both mechanisms are economically equivalent – a tax on regular participants paid to those with an execution advantage. But in TradFi the mechanism is largely invisible, estimated through academic modelling and largely unaddressed by regulation despite being well understood. In DeFi, every MEV transaction is recorded on-chain, publicly attributable, and measurable in real time. The transparency that DeFi's architecture enforces as a general principle applies equally to its dysfunctions. Whether that visibility translates into better outcomes for regular participants depends on governance responses that remain, at the time of writing, unresolved.

Liquidity fragmentation presents a parallel contrast. In US equity markets, Reg NMS mandates that brokers route orders to the venue offering the best available price, creating a legally enforced integration of liquidity across fragmented venues. DeFi has no such mandate: the same asset routinely trades at different prices on Uniswap, Curve, and Hyperliquid

simultaneously, with no obligation on any party to route to the best price. DEX aggregators such as 1inch perform a functionally similar role – routing across venues to minimise execution cost – but as a profit-motivated service rather than a regulatory obligation. The party who bears the fragmentation cost in each system is different: in TradFi it is the broker who bears the routing obligation; in DeFi it is the trader who must know to use an aggregator.

CHAPTER 5

FINDINGS AND DISCUSSION

5.1 Principal

Findings

The analysis in Chapter 4 produces five findings that collectively answer the research question: how do market microstructure properties differ across TradFi, CEX, AMM-DEX, and on-chain CLOB venues, and do standard quantitative tools remain valid across them?

Finding 1: Regulation does not automatically confer execution quality advantage. NYSE spreads for SPY are the highest of the four venues at every trade size, exceeding both CEX and on-chain CLOB venues by two orders of magnitude at retail trade sizes. This is not a failure of the regulatory framework – Reg NMS is designed to ensure best execution across fragmented venues, not to minimise single-venue spread – but it does challenge the assumption that TradFi infrastructure is uniformly superior on execution cost grounds. For a retail investor executing a single transaction on a single venue, a Binance or Hyperliquid account offers materially lower transaction costs than the NYSE equivalent.

Finding 2: AMM and CLOB architectures occupy different cost regimes that do not simply converge. Uniswap v3's concentrated liquidity mechanism produces a flat spread profile determined entirely by the LP fee tier, while CLOB venues produce near-zero spreads at small sizes that rise with trade size. Neither architecture dominates unconditionally: the AMM's fee floor is a ceiling that CLOBs undercut at retail sizes but approach at institutional sizes. The practical implication is that optimal venue selection depends on trade size and cannot be determined by architecture alone.

Finding 3: The on-chain CLOB achieves functional parity with CEX across all measured dimensions. Hyperliquid's spread profile, funding rate behaviour, and volatility properties track Binance closely throughout the sample period. This is a significant finding because prior literature treated on-chain execution as structurally inferior to CEX execution on latency and depth grounds. The data suggests that, at least for liquid assets and standard trade sizes, the infrastructure gap has closed.

Finding 4: Standard quantitative tools require structural adjustment before cross-venue application. The Amihud ratio is not a valid illiquidity measure for AMM pools due to the endogeneity of arbitrage volume. GARCH-based volatility targeting calibrated on equity regime persistence will systematically misprice crypto risk due to the higher frequency and different distribution of volatility regimes. ACF-based mean-reversion signals validated on SPY will encounter a structurally different autocorrelation environment in BTC and ETH. In each case, the issue is not that the tool is wrong – it is that the tool embeds assumptions about market structure that do not hold uniformly across venue types.

Finding 5: HIP-3 synthetic equity tokens introduce a qualitatively new category of oracle risk with regulatory implications that existing frameworks are not designed to handle. The TSLA deviation analysis demonstrates that the synthetic token price is structurally unanchored during NYSE-closed hours, with observed deviations of 5–10% over multi-day windows. This is not a temporary inefficiency that arbitrage will eliminate – it is a permanent structural feature of any instrument that trades continuously against an underlying that does not. The regulatory question is similarly structural: HIP-3 tokens are neither securities nor derivatives in any existing legal category, and the Howey Test analysis in Chapter 4 suggests that this gap is a design choice rather than an oversight.

5.2 Discussion

These findings have implications that extend beyond the specific venues and assets examined. The most significant is that the traditional mapping between regulation and market quality is breaking down at the execution level. Regulated venues remain superior on investor protection, systemic risk management, and legal recourse – dimensions this paper does not measure. But on the dimensions that directly affect trading cost and price discovery efficiency, the gap between regulated and unregulated venues has narrowed to the point where it can no longer be assumed without measurement.

The convergence of Hyperliquid and Binance suggests that the DeFi-CeFi (decentralised finance–centralised finance) distinction, while architecturally real, may be less meaningful to a trader than the CLOB-AMM distinction. Two venues can share an order book architecture and produce near-identical execution statistics while differing fundamentally on custody model and failure mode – and the analytical tools appropriate for each are different.

The HIP-3 finding raises a broader question about the direction of DeFi development. The original promise of decentralised finance was disintermediation: removing rent-extracting middlemen between savers and borrowers, buyers and sellers. HIP-3 does something different – it recreates exposure to regulated assets inside an unregulated wrapper, preserving returns while stripping the protections regulation attaches to them. Whether this constitutes genuine financial innovation or regulatory arbitrage depends on the observer’s prior, but the data makes the mechanism precise: the synthetic’s price is anchored to the underlying only when arbitrage is mechanically possible, and free to diverge whenever it is not.

The quantitative tools finding points toward a practical agenda. The problem is not that researchers and practitioners use the wrong models – it is that they use correct models with incorrect domain assumptions. GARCH is a valid volatility model; the issue is calibrating its persistence parameter on equity data and applying it to crypto. The Amihud ratio is a valid illiquidity measure; the issue is applying it to a market where volume is partly mechanically generated rather than organic. Developing domain-adjusted versions of standard tools – rather than replacing them entirely – is a more tractable research programme.

CHAPTER 6

CONCLUSIONS

6.1 Summary

This paper set out to examine whether DeFi markets have matured to the point where they can be evaluated on the same microstructure terms as their TradFi counterparts, and whether the quantitative tools developed for traditional markets transfer cleanly across the divide. The answer to both questions is qualified: yes, with important structural caveats that have practical consequences for traders, researchers, and regulators alike.

Across four venue types – NYSE, Binance CEX, Uniswap v3 AMM, and Hyperliquid on-chain CLOB – the data reveals that execution cost is not correlated with regulatory status in the direction most observers would expect. The most regulated venue is the most expensive at every trade size. The least regulated – an on-chain order book with no central operator – matches the CEX on spread, funding rate behaviour, and volatility properties. This does not mean regulation is irrelevant; it means that the costs regulation imposes on execution quality are real and measurable, and that the protections it provides operate on dimensions this paper does not capture: legal recourse, custody safety, and systemic risk containment.

The AMM-CLOB comparison produces a more nuanced finding. Neither architecture dominates unconditionally. AMMs provide a predictable, size-invariant fee structure that traders can model precisely; CLOBs provide near-zero retail spreads that rise with size. The choice between them is a function of trade size and whether the trader values cost predictability over cost minimisation.

The HIP framework analysis identifies a new category of financial instrument that sits outside existing regulatory definitions and introduces structural risks – oracle dependency, after-hours price dislocation, and layered synthetic exposure – that are not adequately captured by frameworks developed for either traditional derivatives or earlier DeFi primitives.

6.2 Limitations

Several limitations constrain the generalisability of these findings. First, the microstructure data is a snapshot: order book depth and spread measurements reflect a single point in time and may not represent average conditions across the full sample period. Second, the HIP-3 analysis is limited to a single asset (TSLA) over a 45-day window, which is insufficient to characterise the full distribution of oracle deviations across different market conditions. Third, the paper does not measure the investor protection dimensions of regulation – recourse mechanisms, disclosure requirements, and systemic oversight – which may justify execution cost differences that spread data alone cannot explain.

6.3 Future

Research

Three directions follow directly from the findings. First, a longitudinal spread study across multiple assets and time periods would determine whether Hyperliquid’s execution parity with Binance is persistent or a feature of current market structure. Second, a systematic analysis of HIP-3 oracle deviations across multiple assets and conditions – particularly around corporate events such as earnings releases, which occur outside NYSE hours – would quantify the tail risk that the TSLA window analysis suggests is material. Third, developing domain-adjusted versions of the Amihud ratio and GARCH models that account for structural properties of AMM and on-chain CLOB markets would produce more reliable tools for the growing community of researchers and practitioners working across venue types.

The broader implication is that the boundary between TradFi and DeFi is becoming a productive object of study rather than a dividing line between two separate literatures. The markets are converging faster than the analytical frameworks used to study them, and closing that gap is both tractable and overdue.

REFERENCES

- Aigner, A. A., & Dhaliwal, G. (2021). *UNISWAP: Impermanent loss and risk profile of a liquidity provider* (Tech. Rep. No. 2106.14404). arXiv. (arXiv preprint)
- Alexander, C., Deng, J., Feng, J., & Wan, H. (2023). Net buying pressure and the information in Bitcoin option trades. *Journal of Financial Markets*, 63, 100764. doi: doi:10.1016/j.finmar.2022.100764
- Amihud, Y. (2002). Illiquidity and stock returns: Cross-section and time-series effects. *Journal of Financial Markets*, 5(1), 31–56. doi: doi:10.1016/S1386-4181(01)00024-6
- Aspris, A., Foley, S., Svec, J., & Wang, L. (2021). Decentralized exchanges: The “wild west” of cryptocurrency trading. *International Review of Financial Analysis*, 77, 101845. doi: doi:10.1016/j.irfa.2021.101845
- Barbon, A., & Rinaldo, A. (2022). *On the quality of cryptocurrency markets: Centralized versus decentralized exchanges* (Tech. Rep. No. 2112.07386). arXiv. (arXiv preprint)
- Capponi, A., & Jia, R. (2021). *The adoption of blockchain-based decentralized exchanges* (Tech. Rep. No. 2112.06810). arXiv. (arXiv preprint)
- DappRadar. (2025). *Defi industry report 2025*. <https://dappradar.com>. (Accessed March 2026)
- Del Monte, G., de Lucio, J., & Urban, J. (2025). Current understanding of impermanent loss risk in automated market makers. *Blockchain: Research and Applications*. (In press)
- Glosten, L. R., & Milgrom, P. R. (1985). Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics*, 14(1), 71–100. doi: doi:10.1016/0304-405X(85)90044-3
- Heimbach, L., Wang, Y., & Wattenhofer, R. (2022). *Behavior of liquidity providers in decentralized exchanges* (Tech. Rep. No. 2105.13822). arXiv. (arXiv preprint)
- Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica*, 53(6), 1315–1335. doi: doi:10.2307/1913210
- Milionis, J., Moallemi, C. C., Roughgarden, T., & Zhang, A. L. (2022). *Automated market making and loss-versus-rebalancing* (Tech. Rep. No. 2208.06046). arXiv. (arXiv preprint)
- NFTPlazas. (2025). *Decentralised perpetuals volume report 2025*. <https://nftplazas.com>. (Accessed March 2026)
- Park, A. (2025). Conceptualizing decentralized finance. *Journal of Financial Regulation*. (Advance access)
- Schär, F. (2021). Decentralized finance: On blockchain- and smart contract-based financial markets. *Federal Reserve Bank of St. Louis Review*, 103(2), 153–174. doi: doi:10.20955/r.103.153-74
- Werner, S. M., Perez, D., Gudgeon, L., Klages-Mundt, A., Harz, D., & Knottenbelt, W. J. (2022).

- SoK: Decentralized finance (DeFi). In *Proceedings of the 4th acm conference on advances in financial technologies*. doi: doi:10.1145/3558535.3559780
- Wu, X., Deng, W., Quan, Y., & Zhang, L. (2024). *Trust dynamics and market behavior in cryptocurrency: A comparative study of centralized and decentralized exchanges* (Tech. Rep. No. 2404.17227). arXiv. (arXiv preprint)
- Xu, J., & Vadgama, N. (2022). From banks to DeFi: The evolution of the lending market. In *Enabling the internet of value*. Springer. doi: doi:10.1007/978-3-031-18679-0

APPENDIX A

DATA SOURCES AND VARIABLE DEFINITIONS

A.1 Data	Sources	by	Figure
----------	---------	----	--------

Figure 1 – Four-Venue Effective Spread. NYSE SPY bid-ask spread data is sourced from yfinance using daily OHLCV (Open, High, Low, Close, Volume) records; the effective spread is approximated as the daily high-low range divided by the closing mid-price. Binance BTC/USDT order book snapshots are collected via the Binance REST API through the CCXT library at ten trade-size increments between \$1,000 and \$1,000,000. Uniswap v3 USDC/ETH pool data is retrieved from the Alchemy Ethereum RPC endpoint using direct calls to the pool contract; the effective spread equals the 0.05% LP fee tier, as price impact is negligible at these trade sizes for this pool. Hyperliquid BTC order book data is sourced from the Hyperliquid public REST API. All snapshots were collected in March 2026.

Figure 2 – CME Basis vs Perpetual Funding Rates. CME Bitcoin futures prices are sourced from yfinance using the BTC=F ticker (front-month contract). The annualised basis is computed as $\text{Basis} = [(F/S) - 1] \times (365/\text{DTE})$, where F is the futures price, S is the spot price from Binance, and DTE is days to expiry. Binance perpetual funding rates are sourced via the Binance REST API (/fapi/v1/fundingRate endpoint). Hyperliquid funding rates are sourced from the Hyperliquid public API (/info endpoint, fundingHistory method). Sample period: January 2022 to December 2025.

Figure 3 – IV Skew Proxies. The SPX skew proxy uses the CBOE Skew Index, downloaded from the CBOE public data portal. The BTC skew proxy is constructed from the Deribit Volatility Index (DVOL) using a z-score normalisation over a 90-day rolling window. Both series are smoothed with a 30-day rolling mean. Vertical event markers are placed at: FTX collapse (11 November 2022), spot Bitcoin ETF approval (10 January 2024), and BTC all-time high (14 March 2024).

Figure 4 – HIP-3 Oracle Deviation (TSLA). Synthetic TSLA token price data is sourced from the Hyperliquid spot candlestick API using market identifier @264 at one-hour resolution. NYSE reference prices are sourced from yfinance using the TSLA ticker at one-hour resolution and forward-filled during market-closed periods. Timestamps are rounded to the nearest hour to align the two series. NYSE closed periods are identified by checking that the timestamp falls within 13:30–20:00 UTC on a weekday that is not a US public holiday. The observation window is 6 January to 20 February 2026, beginning seven days after the token’s listing date of 30 December 2025.

Figures 5 & 6 – GARCH Volatility and ACF. Daily close-to-close log returns are computed as $r_t = \ln(P_t/P_{t-1}) \times 100$ for SPY, BTC-USD, and ETH-USD using yfinance price data. The

GARCH(1,1) model is estimated using the arch Python package with a normal error distribution. High-volatility regimes are defined as periods where conditional volatility exceeds the 75th percentile of the full-sample distribution. ACF values at lags 1–20 are computed using `statsmodels.tsa.stattools.acf` with the FFT method; 95% confidence bounds are calculated as $\pm 1.96/\sqrt{n}$, where n is the number of observations. Sample period: January 2022 to December 2025.

A.2 Variable

Definitions

Effective spread The cost of a round-trip transaction expressed as a percentage of trade value, measured as twice the distance between the transaction price and the prevailing mid-quote.

Amihud illiquidity ratio The daily ratio of absolute return to dollar volume: $ILLIQ_t = |r_t|/V_t$, where r_t is the daily return and V_t is daily dollar volume. A higher value indicates lower liquidity.

Annualised CME basis The annualised premium of the CME front-month Bitcoin futures price over the Binance spot price: $[(F/S) - 1] \times (365/\text{DTE})$.

Perpetual funding rate A periodic payment (typically every 8 hours) between long and short positions in a perpetual swap contract, calibrated to keep the contract price close to the underlying spot price. A positive rate means longs pay shorts; a negative rate means shorts pay longs.

25-delta risk reversal The implied volatility of a 25-delta call option minus the implied volatility of a 25-delta put option at the same expiry. A negative value indicates that put options are more expensive than calls (downside skew); a positive value indicates the reverse.

Oracle deviation The percentage difference between a synthetic token’s market price on Hyperliquid and the reference price of its underlying asset: $\delta_t = (P_t^{\text{synth}}/P_t^{\text{ref}} - 1) \times 100$.

Conditional volatility (GARCH) The time-varying volatility estimate produced by a GARCH(1,1) model: $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$, where ω is the long-run variance, α captures the ARCH effect (sensitivity to recent shocks), and β captures persistence.

Autocorrelation at lag k The correlation between r_t and r_{t-k} , measuring the degree to which past returns predict future returns at horizon k .

A.3 Software

and

Reproducibility

All data collection, processing, and figure generation was performed in Python 3.11. Key libraries used are: `yfinance` (market data retrieval), `ccxt` (Binance API interface), `requests` (Hyperliquid and Deribit API calls), `arch` (GARCH estimation), `statsmodels` (ACF computation), `pandas` and `numpy` (data processing), and `matplotlib` (figure generation). All figures were generated from live API data and are reproducible given equivalent API access at the time of data collection. The Hyperliquid TSLA synthetic token (@264) was settled and delisted in March 2026; historical candlestick data remains available via the Hyperliquid archive

endpoint.